60 DAYS | QUALITY ENGINEERING LAB

FROM THEORY TO REAL-WORLD
INDUSTRIAL APPLICATIONS

DAY 05 | EXPERIMENT NO. 05

INVENTORY OPTIMIZATION USING ABC ANALYSIS, EOQ, REORDER POINT & SAFETY STOCK

APPLICATION IN A MANUFACTURING ORGANIZATION

PREPARED BY
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2. Experiment Information

Field	Details
Experiment	Application of ABC Analysis, EOQ, Reorder Point and Safety Stock for Inventory Optimization
Experiment number	05
Project	60 Days | Quality Engineering Lab
Prepared by	Jagriti Singh
Role	Quality Engineer
Domain	Quality Engineering, Inventory Management, Industrial Engineering, Production Planning
Dataset status	Fictional company and simulated data created only for educational analysis

3. Abstract

This experiment applies four practical inventory-control methods to a simulated manufacturing organization: ABC classification, Economic Order Quantity (EOQ), Reorder Point (ROP) and Safety Stock. A consistent 20-item dataset covers ten production materials and ten maintenance spares. Annual consumption value is used to rank and classify items. EOQ is calculated using annual demand, per-order procurement cost and annual holding cost per unit. Reorder points use average daily demand, supplier lead time and a statistically estimated safety stock. A modeled before-and-after scenario compares inventory investment and annual relevant inventory costs. The analysis demonstrates how differentiated controls can protect production continuity while reducing excess stock. All results are model outputs, not measured industrial performance.

4. Introduction

Inventory is the material held to support production, maintenance and dispatch. Too little inventory can stop a line; too much ties up cash, occupies floor space, increases handling and can create deterioration or obsolescence. Inventory optimization is therefore a balance between service and cost, not simply a target to minimize stock.

The experiment uses value-based prioritization (ABC), an economic lot-size calculation (EOQ), a time-phased trigger (ROP) and a buffer against uncertainty (safety stock). The methods are complementary: ABC decides management attention, EOQ informs order size, and ROP plus safety stock help decide when to order.

5. Experiment Objectives

• Build and validate one consistent 20-item inventory dataset.

• Classify materials using annual consumption value and cumulative contribution.

• Calculate EOQ and its annual ordering and cycle-stock holding costs.

• Estimate safety stock and reorder points using explicit demand-variability assumptions.

• Evaluate inventory investment, relevant cost and turnover.

• Compare a defined existing-stock model with a proposed control model.

• Translate the calculations into actionable Quality Engineering controls.

6. Industrial Background

A discrete manufacturing plant typically receives metals, polymers, bought-out components, packaging and consumables. Maintenance teams also hold bearings, seals, valves, electrical modules and filters. These items differ substantially in annual spend, criticality, consumption pattern and replenishment lead time. A single review frequency or reorder rule is therefore inefficient.

The scenario assumes periodic purchasing, inconsistent min/max levels, limited cycle counting and manual escalation of shortages. The modeled improvement introduces ABC-based review, EOQ lot sizing, calculated reorder triggers, supplier lead-time monitoring and more reliable records.

7. Hypothetical Company Profile

Apex Motion Components Pvt. Ltd. (fictional)

Apex Motion Components is a hypothetical medium-sized manufacturer of industrial pump assemblies, motor mounting brackets and compact material-handling modules. Its processes include sheet cutting, turning, milling, welding, surface finishing, assembly and functional testing. The plant operates one main production shift with planned overtime during peak demand.

Products: pump assemblies, motor brackets and conveyor drive modules. **Production materials:** steel sheet, aluminium extrusion, copper wire, stainless bar, polymer, fasteners, paint, cartons and cutting inserts. **Maintenance spares:** bearings, hydraulic seals, solenoid valves, belts, PLC modules, pumps, switches, oil, contactors and air filters.

Current practices assumed: purchase requests are raised from manual stock checks; supplier lead times are stored in spreadsheets; reorder levels are not consistently recalculated; spare parts are sometimes treated as ordinary consumables; and cycle counts are irregular.

Observed scenario problems: excess slow-moving stock, emergency purchases, shortages of selected critical spares, uneven storage utilization, uncertain stock records and production interruptions. These are scenario conditions created for the experiment, not claims about a real company.

8. Technical Background

8.1 Inventory management

Inventory management controls what to stock, how much to stock, where it is stored and when replenishment should occur. Good control supports uninterrupted production, traceability, working-capital discipline and lower total cost. It also requires accurate receipts, issues, returns, scrap and physical counts.

8.2 ABC analysis

ABC analysis ranks items by annual consumption value = annual demand × unit purchase cost. Category A items account for the largest share of value and receive tight review; B items receive normal periodic control; C items are low-value and can use simpler controls. The conventional cumulative-value bands used here are approximately A: first 80%, B: next 15% (to 95%), C: remaining 5%. Classification is a management heuristic, not a measure of operational criticality.

8.3 Economic Order Quantity

EOQ estimates the order quantity that balances annual ordering cost against annual cycle-stock holding cost. EOQ = √(2DS/H), where D is annual demand (units/year), S is ordering cost (■/order), and H is annual holding cost (■/unit/year). The calculation assumes stable demand, fixed ordering cost, constant unit price, replenishment without quantity discounts and no capacity constraint. Safety stock is handled separately.

8.4 Reorder Point

ROP is the inventory position at which a replenishment order should be initiated. ROP = average daily demand × lead time + safety stock. Use inventory position (on-hand + on-order − committed/backorders), not only the physical shelf balance, when applying the trigger.

8.5 Safety stock

Safety stock is a buffer against demand variation and/or lead-time variation. For this experiment, daily demand standard deviation is assumed to be 20% of average daily demand, lead time is treated as fixed, and a service factor z = 1.65 (approximately 95% one-sided cycle-service target under

a normal-demand approximation) is used. Safety stock = z × daily demand standard deviation × √(lead time in days). This is a simplified statistical method; actual plants should estimate variability from transaction history and include lead-time variation when material.

8.6 Inventory turnover

Inventory turnover = cost of goods sold (COGS) ÷ average inventory value. It indicates how many times the average inventory investment is consumed or sold over a period. Very low turnover may indicate excess stock; very high turnover can indicate lean stock or shortage exposure. It must be interpreted with service level and product mix.

9. Experimental Methodology

• Define the hypothetical plant, item list and annual demand assumptions.

• Calculate annual consumption value for each item and rank items in descending order.

• Calculate value share and cumulative share; assign A/B/C classes using the stated bands.

• Calculate EOQ for every item using the same demand, ordering cost and holding-rate assumptions; discuss selected items in detail.

• Estimate daily demand as annual demand/365; calculate safety stock and ROP using fixed lead time and the common variability rule.

• Model existing average stock as 90 days of demand. Model proposed average stock as EOQ/2 plus safety stock.

• Compare holding and ordering costs; calculate turnover using a stated COGS proxy; evaluate modeled service KPIs separately.

Unit convention: demand and order quantities are in units; lead time is calendar days; costs are Indian rupees (■); annual rates cover 365 days. EOQ values are rounded for operational use, but cost calculations use unrounded EOQ.

10. Experimental Dataset

The following master dataset is the single source used throughout the experiment. Annual holding cost per unit is set at 20% of unit purchase cost for every item. Ordering cost is a fixed administrative/procurement cost per purchase order. Lead times are assumed supplier calendar days.

ID	Material description	Type	Annual demand (units)	Unit cost (■)	Annual value (■)	Lead time (days)	Order cost (■/order)	Holding cost (■/unit/yr)
RM01	Mild steel sheet 2 mm	Raw material	12,000	82.00	984,000	7	900	16.40
RM02	Aluminium extrusion	Raw material	4,200	245.00	1,029,000	10	1,100	49.00
RM03	Copper winding wire	Raw material	1,800	680.00	1,224,000	14	1,250	136.00

ID	Material description	Type	Annual demand (units)	Unit cost (■)	Annual value (■)	Lead time (days)	Order cost (■/order)	Holding cost (■/unit/yr)
RM04	Stainless steel bar	Raw material	2,600	310.00	806,000	9	1,000	62.00
RM05	Polymer granules	Raw material	9,000	96.00	864,000	6	850	19.20
RM06	Bearing steel tube	Raw material	1,500	430.00	645,000	12	1,100	86.00
RM07	Fastener set M8	Raw material	24,000	4.50	108,000	5	450	0.90
RM08	Industrial paint	Raw material	1,200	185.00	222,000	8	700	37.00
RM09	Packaging carton	Raw material	18,000	12.00	216,000	4	500	2.40
RM10	Cutting tool inserts	Raw material	3,600	58.00	208,800	7	650	11.60
SP01	Motor bearing 6205	Spare part	240	420.00	100,800	12	900	84.00
SP02	Hydraulic seal kit	Spare part	120	1,350.00	162,000	18	1,100	270.00
SP03	Pneumatic solenoid valve	Spare part	48	2,850.00	136,800	21	1,300	570.00
SP04	V-belt set	Spare part	180	310.00	55,800	8	650	62.00
SP05	PLC input module	Spare part	12	18,500.00	222,000	30	1,500	3,700.00
SP06	Cooling pump	Spare part	8	24,000.00	192,000	35	1,600	4,800.00
SP07	Limit switch	Spare part	60	950.00	57,000	10	700	190.00
SP08	Gearbox oil 20 L	Spare part	96	2,200.00	211,200	7	600	440.00
SP09	Contactor 32 A	Spare part	36	1,750.00	63,000	14	750	350.00
SP10	Air filter element	Spare part	144	520.00	74,880	9	600	104.00

Total annual consumption value: ■7,582,280 across 20 items. The total is annual usage value, not the value of stock physically on hand.

11. ABC Analysis

Annual consumption value is calculated item by item. Total annual consumption value is the denominator for value percentage. Items are ranked from highest to lowest annual value; the cumulative percentage determines the ABC class.

Rank	Item	Annual value (■)	Share (%)	Cumulative (%)	Class
1	RM03	1,224,000	16.14	16.14	A
2	RM02	1,029,000	13.57	29.71	A

Rank	Item	Annual value (■)	Share (%)	Cumulative (%)	Class
3	RM01	984,000	12.98	42.69	A
4	RM05	864,000	11.39	54.09	A
5	RM04	806,000	10.63	64.72	A
6	RM06	645,000	8.51	73.22	A
7	RM08	222,000	2.93	76.15	A
8	SP05	222,000	2.93	79.08	A
9	RM09	216,000	2.85	81.93	A
10	SP08	211,200	2.79	84.71	B
11	RM10	208,800	2.75	87.47	B
12	SP06	192,000	2.53	90.00	B
13	SP02	162,000	2.14	92.14	B
14	SP03	136,800	1.80	93.94	B
15	RM07	108,000	1.42	95.36	B
16	SP01	100,800	1.33	96.69	C
17	SP10	74,880	0.99	97.68	C
18	SP09	63,000	0.83	98.51	C
19	SP07	57,000	0.75	99.26	C
20	SP04	55,800	0.74	100.00	C

Classification result: Category A: 9 items, 81.9% of annual value, Category B: 6 items, 13.4% of annual value, Category C: 5 items, 4.6% of annual value. The largest annual-value items should receive frequent review, approved suppliers, tighter record accuracy and management attention. ABC alone must not downgrade a low-value but production-critical spare.

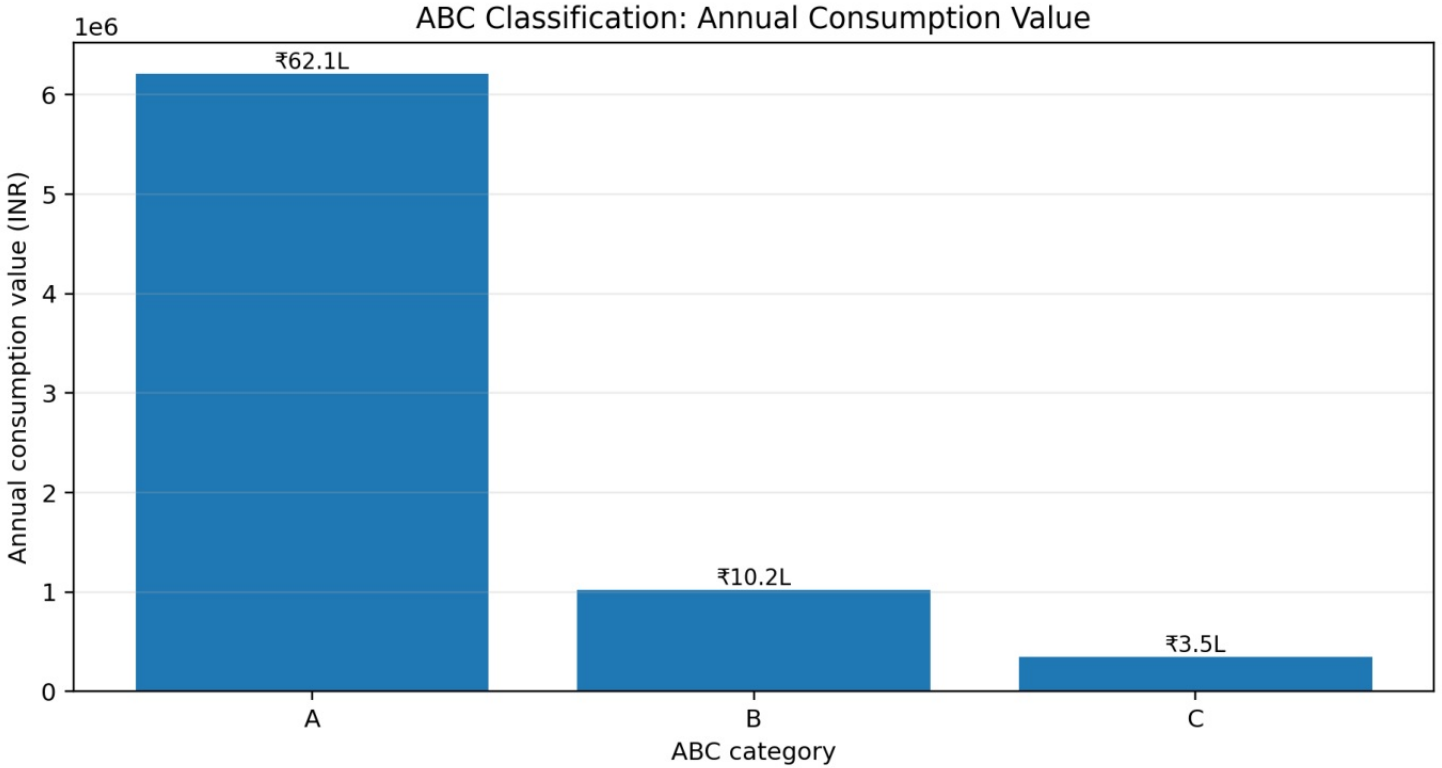


Figure 1. ABC category contribution to total annual consumption value (■).

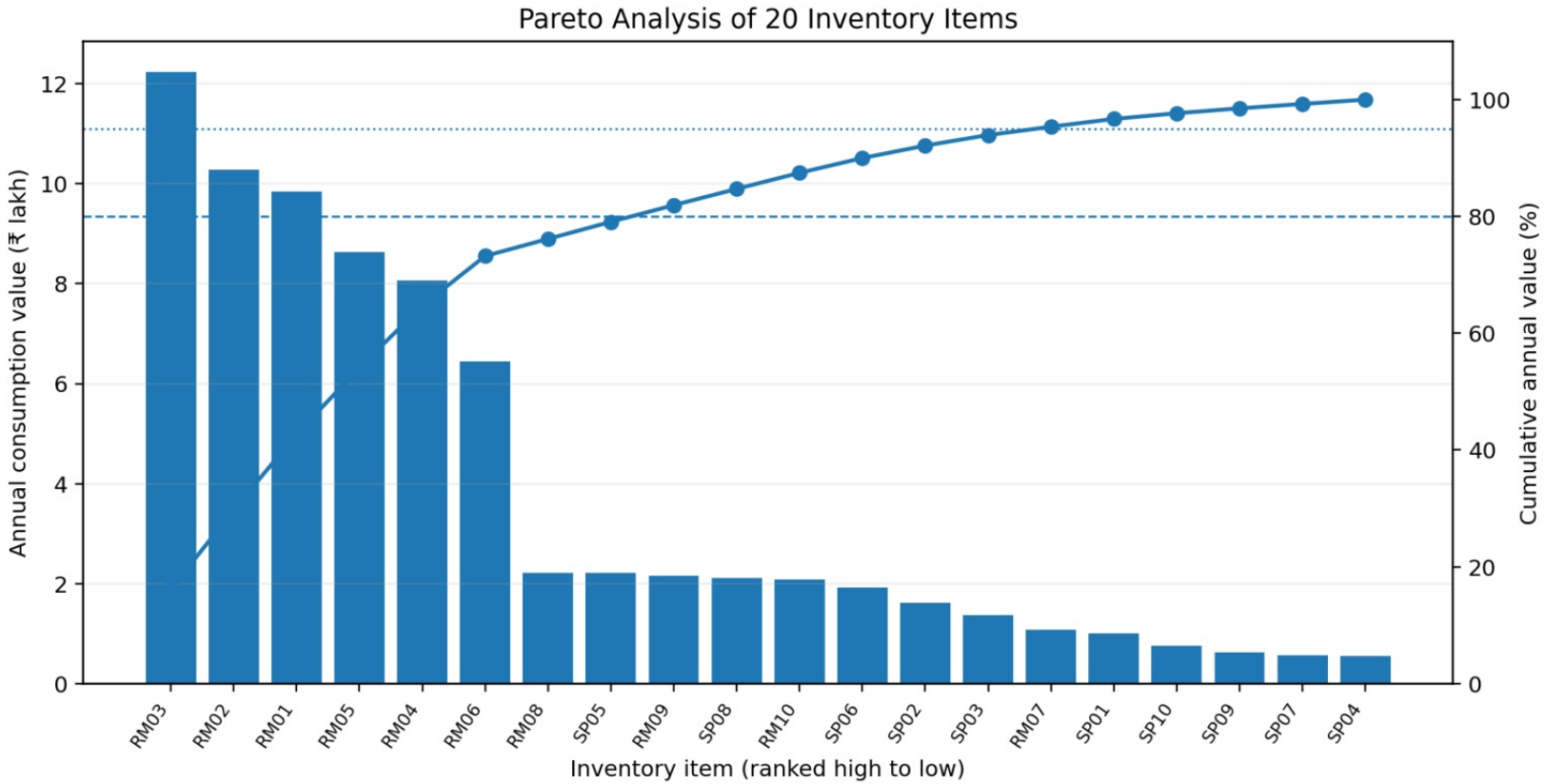


Figure 2. Pareto chart: bars show annual value by ranked item; line shows cumulative percentage. Dashed reference levels indicate 80% and 95%.

12. EOQ Analysis

EOQ is calculated for all 20 items in the workbook. The following four examples cover high-value production materials and one critical, high-unit-cost spare. H = 20% × unit cost. Annual ordering cost = (D/EOQ) × S. Annual cycle-stock holding cost = (EOQ/2) × H.

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EOQ is calculated for all 20 items in the workbook. The following four examples cover high-value production materials and one critical, high-unit-cost spare. H = 20% × unit cost. Annual ordering cost = (D/EOQ) × S. Annual cycle-stock holding cost = (EOQ/2) × H.

RM01 — Mild steel sheet 2 mm

Given: D = 12,000 units/year; S = ■900/order; unit cost = ■82.00; H = 20% × ■82.00 = ■16.40/unit/year.

Formula and substitution: EOQ = √(2DS/H) = √[2 × 12,000 × 900 ÷ 16.40] = **1,147.6 units** (operational suggestion: 1148 units).

Orders/year: D/EOQ = 12,000/1,147.6 = 10.46. **Annual ordering cost:** 10.46 × ■900 = ■9,411. **Annual cycle-stock holding cost:** (EOQ/2) × H = (1,147.6/2) × ■16.40 = ■9,411.

Engineering interpretation: under the EOQ assumptions, this lot size balances the two relevant cost components. It does not include safety stock, purchase expenditure, freight discounts or capacity constraints.

RM02 — Aluminium extrusion

Given: D = 4,200 units/year; S = ■1,100/order; unit cost = ■245.00; H = 20% × ■245.00 = ■49.00/unit/year.

Formula and substitution: EOQ = √(2DS/H) = √[2 × 4,200 × 1,100 ÷ 49.00] = **434.2 units** (operational suggestion: 435 units).

Orders/year: D/EOQ = 4,200/434.2 = 9.67. **Annual ordering cost:** 9.67 × ■1,100 = ■10,639. **Annual cycle-stock holding cost:** (EOQ/2) × H = (434.2/2) × ■49.00 = ■10,639.

Engineering interpretation: under the EOQ assumptions, this lot size balances the two relevant cost components. It does not include safety stock, purchase expenditure, freight discounts or capacity constraints.

RM03 — Copper winding wire

Given: D = 1,800 units/year; S = ■1,250/order; unit cost = ■680.00; H = 20% × ■680.00 = ■136.00/unit/year.

Formula and substitution: EOQ = √(2DS/H) = √[2 × 1,800 × 1,250 ÷ 136.00] = **181.9 units** (operational suggestion: 182 units).

Orders/year: D/EOQ = 1,800/181.9 = 9.90. **Annual ordering cost:** 9.90 × ■1,250 = ■12,369. **Annual cycle-stock holding cost:** (EOQ/2) × H = (181.9/2) × ■136.00 = ■12,369.

Engineering interpretation: under the EOQ assumptions, this lot size balances the two relevant cost components. It does not include safety stock, purchase expenditure, freight discounts or capacity constraints.

SP05 — PLC input module

Given: D = 12 units/year; S = ■1,500/order; unit cost = ■18,500.00; H = 20% × ■18,500.00 = ■3,700.00/unit/year.

Formula and substitution: EOQ = √(2DS/H) = √[2 × 12 × 1,500 ÷ 3,700.00] = **3.1 units** (operational suggestion: 4 units).

Orders/year: D/EOQ = 12/3.1 = 3.85. **Annual ordering cost:** 3.85 × ■1,500 = ■5,771. **Annual cycle-stock holding cost:** (EOQ/2) × H = (3.1/2) × ■3,700.00 = ■5,771.

Engineering interpretation: under the EOQ assumptions, this lot size balances the two relevant cost components. It does not include safety stock, purchase expenditure, freight discounts or capacity constraints.

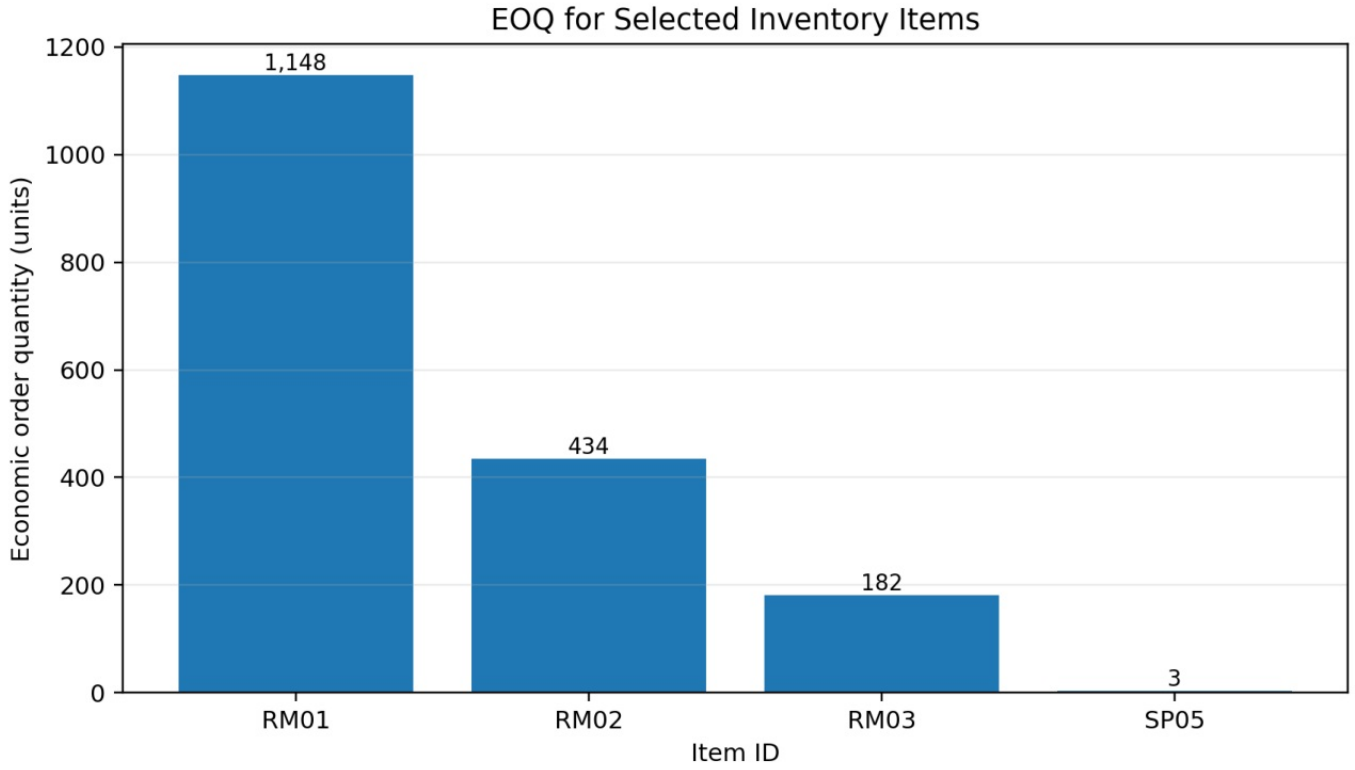


Figure 3. EOQ quantities for the four selected items. Different item scales are expected because annual demand and unit economics differ.

13. Reorder Point Analysis

Average daily demand = annual demand/365. The trigger is ROP = (average daily demand × supplier lead time) + safety stock. The operational reorder point is rounded upward to a whole unit. The calculation assumes continuous monitoring of inventory position.

Item	Daily demand (units/day)	Lead time (days)	Safety stock (units)	ROP exact (units)	ROP trigger (units)
RM01	32.88	7	28.7	258.8	259
RM02	11.51	10	12.0	127.1	128
SP05	0.03	30	0.1	1.0	2

Industrial example: RM01 — Mild steel sheet 2 mm

Daily demand = 12,000/365 = 32.88 units/day. Expected lead-time demand = 32.88 × 7 = 230.14 units. Safety stock = 28.70 units. Therefore ROP = 230.14 + 28.70 = 258.84; trigger at 259 units.

Action: when inventory position reaches 259 units, release or expedite a replenishment order after checking open purchase orders, quality holds and committed demand.

Industrial example: RM02 — Aluminium extrusion

Daily demand = 4,200/365 = 11.51 units/day. Expected lead-time demand = 11.51 × 10 = 115.07 units. Safety stock = 12.01 units. Therefore ROP = 115.07 + 12.01 = 127.08; trigger at 128 units.

Action: when inventory position reaches 128 units, release or expedite a replenishment order after checking open purchase orders, quality holds and committed demand.

Industrial example: SP05 — PLC input module

Daily demand = 12/365 = 0.03 units/day. Expected lead-time demand = 0.03 × 30 = 0.99 units. Safety stock = 0.06 units. Therefore ROP = 0.99 + 0.06 = 1.05; trigger at 2 units.

Action: when inventory position reaches 2 units, release or expedite a replenishment order after checking open purchase orders, quality holds and committed demand.

Ordering too late can create a shortage before replenishment arrives, causing line stoppage, schedule changes or emergency freight. Excessive safety stock raises working capital, storage and obsolescence exposure. A reorder trigger is only reliable when lead-time and stock records are maintained.

14. Safety Stock Analysis

Safety stock method used: SS = z × σd × √L, where z = 1.65, σd = 0.20 × average daily demand, and L is supplier lead time in days. The same formula is applied to all items; there is no hidden item-specific adjustment. The square-root-of-time rule assumes independent daily demand variation and fixed lead time.

Item	Daily demand	Assumed daily SD	Lead time	Safety stock	ROP units
RM01	32.88	6.58	7	28.7	259
RM02	11.51	2.30	10	12.0	128
RM03	4.93	0.99	14	6.1	76
SP05	0.03	0.01	30	0.1	2
SP06	0.02	0.00	35	0.0	1
SP10	0.39	0.08	9	0.4	4

A high lead-time spare can require a larger buffer even when its annual demand is low. The PLC input module and cooling pump also deserve criticality review: low annual usage does not mean low consequence of failure. For real implementation, validate service targets, demand distributions, supplier delivery variation, repairability and equipment redundancy.

15. Inventory Performance Evaluation

15.1 Inventory investment versus annual operating cost

Total annual consumption value is ■7,582,280. This represents the value of material consumed over a year. The modeled existing average on-hand investment is ■1,869,603 (90 days of demand at purchase cost). The proposed average inventory investment is ■552,216 (EOQ/2 cycle stock plus calculated safety stock). These are average inventory asset values, not annual expenses.

Annual holding cost is calculated as average units × H. Annual ordering cost is orders/year × S. Existing-model holding cost = ■373,921; ordering cost = ■73,811; total relevant annual cost = ■447,732. Proposed-model holding cost = ■110,443; ordering cost = ■106,025; total relevant annual cost = ■216,469.

Total relevant cost excludes purchase spend because annual demand and unit prices are assumed unchanged in both systems. It also excludes stockout penalties, quality losses, freight premiums and discount effects. The proposed holding cost includes safety-stock holding cost through proposed average units.

15.2 Inventory turnover

For a consistent illustrative financial proxy, annual COGS is assumed to be 75% of annual consumption value = ■5,686,710. Turnover using average inventory investment as a proxy for average inventory value is 5,686,710/1,869,603 = **3.04 times/year** in the existing model and 5,686,710/552,216 = **10.30 times/year** in the proposed model. In a real business, use audited COGS and time-weighted average inventory at cost.

15.3 Service performance assumptions

The scenario assumes 18 stockout incidents/year and 96.0% material availability in the existing state, compared with modeled targets of 6 incidents/year and 99.2% availability after control changes. These service figures are scenario projections, not outputs mathematically derived from EOQ. They require validation against actual stockout logs and production issue records.

16. Before-and-After Analysis

Metric	Existing model	Proposed model	Change / interpretation
Average inventory investment	■1,869,603	■552,216	70.5% lower
Annual holding cost	■373,921	■110,443	70.5% lower
Annual ordering cost	■73,811	■106,025	-43.6% lower
Total relevant annual cost	■447,732	■216,469	51.7% lower
Inventory turnover	3.04×	10.30×	Higher modeled turnover
Stockout incidents/year	18	6	Modeled reduction; validate
Material availability	96.0%	99.2%	Modeled improvement
Production interruptions/year	12	4	Modeled reduction; validate

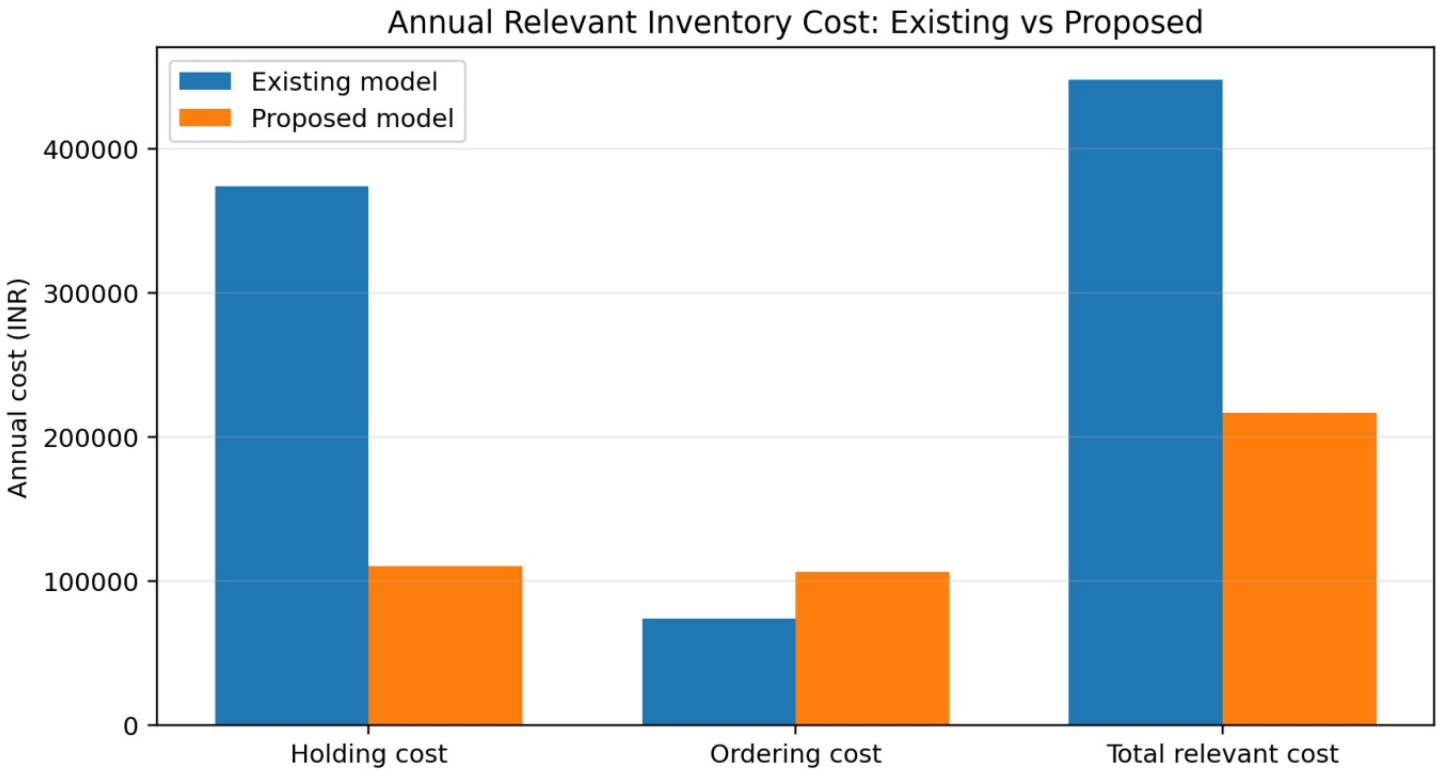


Figure 4. Annual relevant inventory cost comparison. Purchase spend and shortage penalties are excluded.

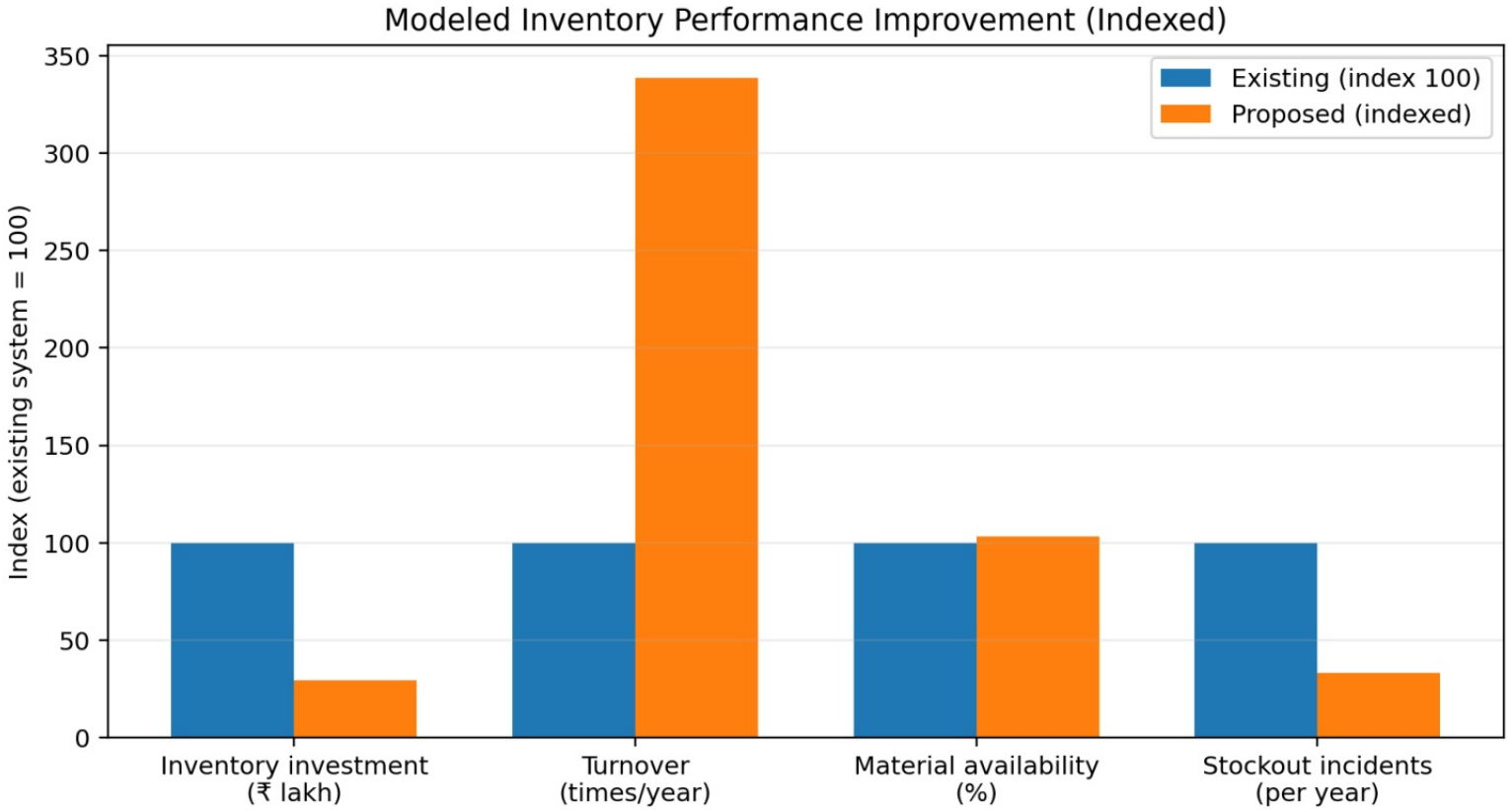


Figure 5. Performance indicators indexed to the existing model (=100). Lower is favorable for inventory investment and stockouts; higher is favorable for turnover and availability. Values are modeled projections.

Projection assumptions

• Existing average stock is set to 90 calendar days of annual demand for every item, representing a simplified excess-stock baseline.

• Proposed average stock is EOQ/2 plus the safety stock calculated using the common 20% daily-demand standard deviation and z=1.65 rule.

• Annual demand, unit prices, ordering cost, holding-rate policy and supplier lead times are held constant between states.

• Existing ordering frequency is approximated as one order every 90 days: annual demand divided by 90-day average stock; proposed ordering frequency is annual demand divided by EOQ.

• COGS is held constant at 75% of annual consumption value solely to illustrate turnover sensitivity.

• Stockout incidents, availability and production interruptions are explicitly assumed scenario KPIs; they are not predicted by the EOQ or ROP equations.

17. Engineering Recommendations

ABC-based control

Review A items weekly and approve changes to their supplier, specification or stock policy; review B items monthly and C items quarterly. Preserve criticality overrides for safety and production-critical spares.

EOQ-based purchasing

Use EOQ as a starting lot-size recommendation. Check MOQ, pack size, shelf life, storage capacity, supplier discounts and cash constraints before releasing a purchase order.

Reorder-level monitoring

Calculate ROP from current demand and supplier lead time. Trigger replenishment using inventory position and an exception dashboard rather than visual shelf checks alone.

Safety stock planning

Estimate demand and lead-time variation from actual history. Set service targets by item criticality and periodically revalidate z-values and buffers.

Supplier lead-time monitoring

Track promised versus actual receipt dates, on-time delivery, lead-time mean and variation. Escalate chronic supplier delays and qualify alternatives for critical items.

Critical spare parts identification

Use equipment criticality, failure consequence, repair lead time, interchangeability and downtime cost alongside ABC value. A low-value part may still be line-critical.

Inventory record accuracy

Use barcode or controlled transaction entry for receipt, issue, return, scrap and transfer. Cycle-count A items frequently and investigate root causes of variance.

Material traceability

Maintain batch/heat/lot identification, certificates, shelf-life controls and status segregation for incoming, accepted, rejected and quarantined material.

Cross-functional coordination

Production supplies the consumption plan; purchasing maintains supplier parameters; stores controls transactions; maintenance identifies critical spares; quality controls acceptance and traceability. Review exceptions together.

18. Results and Discussion

The annual consumption value of the 20 items is ■7,582,280. ABC ranking concentrates value into a small subset of items, supporting differentiated control. EOQ calculations provide an economically balanced lot size under deterministic assumptions. ROP values combine expected lead-time demand with a buffer; this prevents the reorder trigger from relying on average demand alone.

The modeled proposed state reduces average inventory investment from ■1,869,603 to ■552,216 and total relevant annual cost from ■447,732 to ■216,469. These results are sensitive to the 90-day baseline, 20% demand-variation assumption, 20% annual holding rate and fixed supplier lead times. The scenario is a teaching model, not a guaranteed savings forecast.

The most important operational lesson is that inventory control must combine value, uncertainty and criticality. ABC alone cannot protect a low-value critical spare; EOQ alone does not define when to order; and ROP is unreliable if inventory records or lead-time data are poor.

19. Learning Outcomes

- Convert demand and unit price into annual consumption value.
- Rank items and interpret cumulative Pareto contribution.
- Calculate EOQ and separate annual ordering and holding cost.
- Calculate safety stock and ROP with consistent time units.
- Distinguish annual material usage value, inventory investment and annual inventory operating cost.

• Interpret turnover with financial and service context.

• Develop inventory controls from a Quality Engineer’s perspective.

20. Limitations and Assumptions

• All company details, demand, prices, lead times, ordering costs and scenario KPIs are hypothetical.

• Demand is assumed constant and known; seasonality, trend and forecast error are not modeled.

• Lead time is treated as fixed; the safety-stock formula includes demand variation only.

• A 20% holding-cost rate is applied uniformly to unit purchase cost; actual carrying cost may include capital, space, insurance, handling, damage and obsolescence.

• No quantity discounts, minimum order quantities, supplier capacity limits, transport constraints or shelf-life restrictions are modeled.

• The proposed inventory investment is EOQ/2 + safety stock; it is a simplified average-stock approximation.

• The turnover illustration uses COGS = 75% of annual consumption value, not audited accounts.

• Service KPI changes are assumed projections and require pilot validation.

21. Conclusion

The experiment demonstrates a practical framework for inventory optimization in a manufacturing environment. ABC analysis focuses attention on financially significant items; EOQ balances ordering and cycle-stock holding costs; safety stock protects against modeled demand variation; and ROP provides a replenishment trigger tied to lead time. The modeled comparison indicates lower inventory investment and relevant cost under the stated assumptions. For industrial use, the calculations must be calibrated with actual consumption, supplier performance, equipment criticality, financial data and physical stock accuracy.

22. Formula Reference

Measure	Formula	Variable definitions / units
Annual consumption value	D × C	D = annual demand (units/year); C = unit cost (■/unit)
Value percentage	Item annual value ÷ total annual value × 100	Dimensionless (%)

Measure	Formula	Variable definitions / units
Cumulative percentage	Running total of value percentages	Dimensionless (%)
EOQ	√(2DS/H)	D = units/year; S = ■/order; H = ■/unit/year; EOQ = units/order
Orders per year	D ÷ EOQ	orders/year
Annual ordering cost	(D ÷ EOQ) × S	■/year
Cycle-stock holding cost	(EOQ ÷ 2) × H	■/year
Average daily demand	D ÷ 365	units/day
Daily demand SD (assumed)	0.20 × average daily demand	units/day
Safety stock	z × σd × √L	z = service factor; σd = units/day; L = days; result in units
Reorder point	(average daily demand × L) + safety stock	units
Inventory turnover	COGS ÷ average inventory	times/year; both values in ■ at cost
Inventory investment	Average on-hand units × unit cost	■ at cost

23. Numerical Problems with Complete Solutions

Problem 1: ABC value

Question: A component has annual demand 2,400 units and unit cost ■150. Find annual consumption value.
Solution: 2,400 × 150 = ■3,60,000 per year.

Problem 2: ABC share

Question: An item value is ■2,40,000 and total inventory annual value is ■12,00,000. Find share.
Solution: (2,40,000/12,00,000) × 100 = 20%.

Problem 3: Cumulative ABC

Question: Three ranked items contribute 48%, 27% and 15%. Find cumulative share after item 2.
Solution: 48% + 27% = 75%.

Problem 4: EOQ

Question: D=10,000 units/year, S=■500/order, H=■20/unit/year. Find EOQ.
Solution: EOQ=√(2×10,000×500/20)=√500,000=707.1 units; use about 708 units.

Problem 5: Orders/year

Question: D=8,000 units/year and EOQ=400 units. Find orders/year.
Solution: 8,000/400=20 orders/year.

Problem 6: Ordering cost

Question: 20 orders/year at ■650 per order. Find annual ordering cost.
Solution: 20×650=■13,000/year.

Problem 7: Holding cost

Question: EOQ=600 units and H=■30/unit/year. Find annual cycle-stock holding cost.
Solution: (600/2)×30=■9,000/year.

Problem 8: Daily demand

Question: Annual demand is 7,300 units. Find average daily demand using 365 days.
Solution: 7,300/365=20 units/day.

Problem 9: Lead-time demand

Question: Daily demand is 18 units and lead time is 12 days.
Solution: 18×12=216 units.

Problem 10: Safety stock

Question: z=1.65, daily demand SD=3 units, lead time=16 days.
Solution: SS=1.65×3×√16=1.65×3×4=19.8, round up to 20 units.

Problem 11: ROP

Question: Daily demand=25 units/day, lead time=8 days, safety stock=30 units.
Solution: ROP=25×8+30=230 units.

Problem 12: Holding rate

Question: Unit cost=■500 and annual holding rate=18%. Find H.
Solution: H=500×0.18=■90/unit/year.

Problem 13: Turnover

Question: COGS=■48 lakh and average inventory=■8 lakh.
Solution: Turnover=48/8=6 times/year.

Problem 14: Inventory investment

Question: Average stock=750 units and unit cost=■240.
Solution: Investment=750×240=■1,80,000.

Problem 15: Total relevant cost

Question: Annual ordering cost=■14,000 and holding cost=■11,500.
Solution: Relevant cost=14,000+11,500=■25,500/year; excludes purchase spend and shortage cost.

24. Technical Viva Questions with Answers

Viva 1. What is inventory management?

The planning and control of stock quantities, locations, records and replenishment to meet demand at an appropriate total cost.

Viva 2. Why is inventory needed in manufacturing?

To decouple process stages, cover replenishment time and support production and maintenance continuity.

Viva 3. What does ABC stand for?

A value-based classification of inventory into high-, medium- and low-annual-consumption-value groups.

Viva 4. Which formula is used for annual consumption value?

Annual demand multiplied by unit purchase cost.

Viva 5. Does ABC measure criticality?

No. It measures annual usage value; criticality and failure consequence require a separate review.

Viva 6. What is EOQ?

The order quantity that minimizes the sum of modeled annual ordering and cycle-stock holding costs.

Viva 7. What happens if order quantity is much larger than EOQ?

Average cycle stock and holding cost increase, while order frequency and ordering cost decrease.

Viva 8. What happens if order quantity is much smaller than EOQ?

Order frequency and annual ordering cost increase, while cycle-stock holding cost decreases.

Viva 9. What is reorder point?

The inventory-position threshold at which replenishment should be initiated.

Viva 10. Why is safety stock maintained?

To protect against uncertainty in demand and/or replenishment lead time.

Viva 11. What does z represent in the safety-stock equation?

A service factor linked to the desired probability of covering demand during lead time under the assumed distribution.

Viva 12. Why use 365 in daily-demand calculations?

The experiment defines annual demand over 365 calendar days; a real plant should align the denominator with its demand and lead-time calendar.

Viva 13. What is inventory turnover?

COGS divided by average inventory value, expressed as turns per period.

Viva 14. Can a high turnover always be considered good?

No. It may also indicate insufficient buffers and elevated stockout risk.

Viva 15. Why must stock records be accurate for ROP?

An incorrect inventory position can trigger replenishment too early or too late, making the calculated ROP ineffective.

25. Technical References

1. Heizer, J., Render, B. and Munson, C., *Operations Management: Sustainability and Supply Chain Management*, Pearson. Inventory management and EOQ chapters.

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5. APICS/ASCM, *Supply Chain Operations Reference (SCOR) Digital Standard*, for supply-chain process terminology.

6. The numerical data and calculations in this report are an original educational simulation prepared for the 60 Days | Quality Engineering Lab. No external company data are represented.